

Sanghyeok Moon 문상혁

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Research Interests: Efficient AI Systems, AI Agents, Language Models, Retrieval-Augmented Generation

EDUCATION

Mar. 2021 – **Kyung Hee University, Korea, B.S. in Applied Mathematics**
Feb. 2027 (expected) – Double major: Computer Engineering
– GPA: 4.12/4.3 (4.35/4.5)

RESEARCH EXPERIENCE

Mar. 2026 – **Undergraduate Researcher, Efficient Multimodal Intelligence Lab**
Present
Advisor: Prof. Sungha Choi, Kyung Hee University

- Conducted research on memory-agnostic input processing for LLM agent memory systems, focusing on filtering and topic-based chunking across memory backends.

Nov. 2025 – **Undergraduate Researcher, Nonlinear Dynamics Lab**
Jun. 2026
Advisor: Prof. Junpyo Park, Kyung Hee University

- Studied Echo State Networks (reservoir computing method) for time-series prediction.

Jul. 2024 – **Undergraduate Researcher, BioMath Lab**
Aug. 2025
Advisor: Prof. Sunmi Lee, Kyung Hee University

- Developed agent-based models for innovation diffusion and vaccine hesitancy under social and political interactions.
- Constructed large-scale COVID-19 infection networks from 670K+ contact-tracing records and contributed to a network-based R_t estimation method.
- Built SEIR-ABM simulations on multiple network topologies to analyze how outgroup aversion affected vaccine hesitancy.

PUBLICATIONS

International Journals:

- [J1] Lee, Gira; Yoon, KangHyun; Lim, Sang Hun; **Moon, Sanghyeok**; Lee, Sunmi (2026)
“Innovation Diffusion under Political Identity Polarization: A Contextualized Agent-Based Approach”
SAGE Open (SSCI, Q1) Co-author
- [J2] Kim, Byul Nim; Jo, Junwoo; Oh, Chunyoung; **Moon, Sanghyeok**; Abdulali, Arsen; Lee, Sunmi (2025)
“A novel approach to estimating R_t through infection networks: understanding regional transmission dynamics of COVID-19”
Frontiers in Public Health (SCIE, Q1) Co-author

RESEARCH PRESENTATIONS

- | | |
|------|---|
| 2025 | Impacts of Outgroup Aversion and Vaccine Hesitancy on Pandemic Spread
Poster Presentation, JSMB-ASMB (Kyoto, Japan) |
| 2024 | Time-varying Empirical Reproduction Number
Poster Presentation, KSIAM (Busan, Korea) |

PROJECTS

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|------|---|
| 2026 | Reasoning Diversity for Self-Distillation <ul style="list-style-type: none">• Evaluated self-distillation objectives for improving reasoning diversity and mitigating strategy collapse in reinforcement learning for LLMs. |
| 2025 | RAG-based Legal QA Chatbot <ul style="list-style-type: none">• Developed a legal QA chatbot using LangChain, Chroma, and GPT-4o mini, with retrieval-augmented responses grounded in case-law evidence. ABM-based Simulation of Fertility under Wealth Inequality <ul style="list-style-type: none">• Built an agent-based simulation to analyze how wealth inequality and assortative mating affected marriage and fertility dynamics. Price Volatility Forecasting in Daejeon <ul style="list-style-type: none">• Built a forecasting pipeline from 10-year regional price data and applied ARIMA/SARIMA models for price volatility prediction. |

AWARDS

- **Popularity Award**, KHUDA Datathon (KHU'DATA), Kyung Hee University (2024)
- **1st Place**, Big Data & AI Data Analysis Competition, Kyung Hee University (2021)

SKILLS

Programming: Python, SQL

LLM: PyTorch, Transformers, LangChain, Chroma, vLLM

Modeling: Agent-Based Modeling, SEIR Models, ARIMA/SARIMA, Echo State Networks

Tools: NumPy, pandas, scikit-learn, Git, LaTeX

CERTIFICATIONS & LANGUAGE

- TOEIC: 960 / 990 OPIc: IM2
- SQL Developer (SQLD)